Software Applications Development
WorldSkills 2026 Brazil Friendly Competition Online

Utility service

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1. Utility Service API Documentation

In this document, you will find the API endpoints for the Utility Service, which provides functionalities for generating HMAC-based One-Time Passwords (HOTP) and calculating Error Correction Code (ECC) bytes for QR data payloads.

A live API documentation can be found at utility-service.s09.club/documentation.

1.1. HOTP Generation

POST `/hotp`

DESCRIPTION

Generates an HMAC-based One-Time Password (HOTP) code.

REQUEST BODY

Body (JSON):

```
{
  "K": "string", // Raw exactly 10-character secret key
  "C_t": 0 // Counter (non-negative integer)
}
```

RESPONSE: **200 OK**

Body (JSON):

```
{
  "code": "string" // Generated HOTP code
}
```


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1.2. QR Error Correction Code (ECC)

POST `/qr-ecc`

DESCRIPTION

Calculates Error Correction Code (ECC) bytes for a given QR data payload.

REQUEST BODY

Body (JSON):

```
{
  "data_bytes": [0, 1, 2] // Array of 55 data bytes (0-255)
}
```

RESPONSE: **200 OK**

Body (JSON):

```
{
  "ecc_bytes": [0, 1, 2] // Array of generated ECC bytes (0-255)
}
```


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